Diesel Engine Diagnostics Basics: How a diesel engine works.

A diesel engine is a compression-ignition engine. In the four-stroke cycle the piston performs intake, compression, power and exhaust strokes. During the intake stroke only air enters the cylinder.

The air is compressed so strongly, with compression ratios typically between 14:1 and 25:1, that its temperature rises above the autoignition temperature of diesel fuel. Fuel injected near the end of compression ignites by itself, so a diesel engine has no spark plugs. The high compression ratio is the reason diesels are more thermally efficient than petrol engines.

Diesel Engine Diagnostics Basics: Symptoms and what they indicate.

Condition monitoring of a diesel engine relies on a few key measurements: coolant temperature, oil pressure, turbocharger boost pressure, exhaust gas temperature (EGT), crankcase blow-by and vibration.

A high exhaust gas temperature suggests overfuelling or restricted airflow. Low oil pressure at idle points to worn bearings or a failing oil pump. Excessive blow-by indicates worn piston rings or cylinder liners.

The colour of the exhaust smoke is a quick diagnostic clue. Black smoke means incomplete combustion from too much fuel or too little air. White smoke means unburnt fuel or coolant entering the combustion chamber. Blue smoke means that lube oil is being burned, often because of worn valve seals or piston rings.

Diesel Engine Diagnostics Basics: Exhaust aftertreatment.

Modern diesel engines use several systems to reduce emissions. Exhaust gas recirculation (EGR) feeds part of the exhaust back into the intake, which lowers peak combustion temperature and therefore the formation of nitrogen oxides (NOx).

A diesel particulate filter (DPF) traps soot from the exhaust. As soot accumulates the filter must be regenerated by burning the soot off at high temperature; regeneration can be passive, active or forced.

Selective catalytic reduction (SCR) converts NOx into harmless nitrogen and water. It works by injecting diesel exhaust fluid (DEF), a solution of 32.5 percent urea in deionised water, into the exhaust stream upstream of a catalyst. If the DEF tank runs empty, the engine control unit will usually derate the engine.